

Project ID : Group 25-26J-473

1. Topic (12 words max)

Smart Traffic Flow Optimization with Multi Agent Reinforcement Learning

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	-N/A-
Year	-N/A-

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Urban traffic congestion is a growing challenge in modern cities, leading to increased travel time, fuel consumption, and environmental pollution. Traditional traffic signal systems often rely on static timing plans that fail to adapt to real-time traffic dynamics. To address these limitations, our project proposes a smart, adaptive traffic control system based on **Multi-Agent Reinforcement Learning (MART)**. Each intersection is controlled by an intelligent agent capable of learning optimal signal control policies to reduce congestion and delay. The core of our system lies in modeling each traffic light as a Reinforcement Learning (RL) agent that observes its local state—such as vehicle count, queue lengths, signal phases, and neighboring signals. Each agent's actions involve switching signal phases strategically, guided by a reward function that penalizes long wait times and rewards smooth traffic flow. These individual agents are trained under a **centralized training and decentralized execution (CTDE)** paradigm using the **Value Decomposition Network (VDN)** architecture, enabling cooperative learning with minimal inter-agent communication. This allows for scalable deployment while maintaining citywide coordination.

The RL agents are trained and evaluated within a realistic traffic simulation environment built using **SUMO (Simulation of Urban MObility)**. This environment replicates real-world traffic scenarios and integrates real-time emergency detection through sound and video feeds to simulate complex, dynamic conditions. Additionally, the system includes a **Smart Pedestrian Assistance & Safety module**, which enhances the human-centered aspect of urban mobility. This component uses SUMO to simulate pedestrian movement and adapts crossing signals in real-time based on pedestrian presence, behavior, and special needs (such as elderly or differently abled individuals). Features like countdown timers, audio signals, and emergency overrides further ensure safety and inclusivity.

Together, these components form a comprehensive solution to optimize urban traffic flow, improve commuter experience, and increase road safety—paving the way for smarter and more sustainable cities.

References

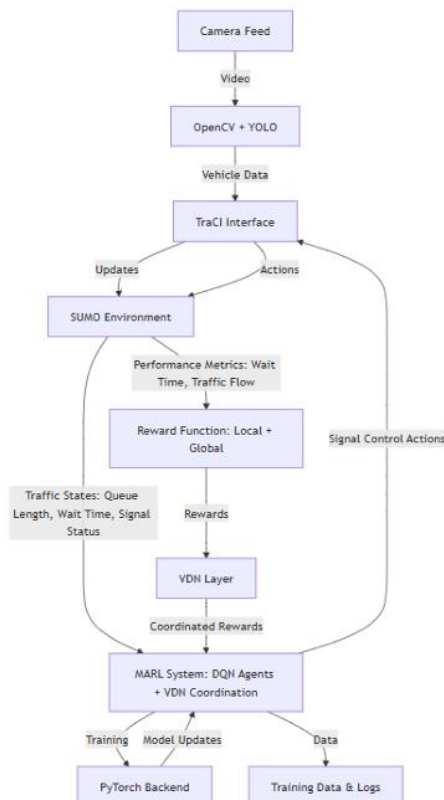
- S. Djahel, D. Smith, and J. Murphy, "A SUMO based evaluation of road incidents' impact on traffic congestion level in smart cities," in Proc. 39th Conf. Local Comput. Netw. Workshops (LCN Workshops), Edmonton, AB, Canada, Sep. 2014, pp. 1–8, doi: 10.1109/LCNW.2014.6927714.
- P. A. Lopez et al., "Microscopic traffic simulation using SUMO," in Proc. 21st IEEE Int. Conf. Intell. Transp. Syst. (ITSC), Maui, HI, USA, Nov. 2018, pp. 2575–2582, doi: 10.1109/ITSC.2018.8569938.
- Z. Cui, K. Henrickson, R. Ke, and Y. Wang, "Traffic graph convolutional recurrent neural network: A deep learning framework for network-scale traffic learning and forecasting," IEEE Trans. Intell. Transp. Syst., vol. 21, no. 11, pp. 4883–4894, Nov. 2020, doi: 10.1109/TITS.2019.2950416.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The proposed system integrates MARL agents with a traffic simulation environment (SUMO) and a real-time vision system to create a smart traffic signal control solution. The system consists of the following components:

- **Traffic State Representation & Local Reward Shaping (Component 1):**
Each traffic agent observes queue length, wait time, and neighbor states. It receives a locally shaped reward that encourages queue reduction and traffic flow improvement.
- **Decentralized MARL Algorithm Implementation (Component 2):**
Using the VDN framework with DQNs, agents are trained in a decentralized manner with partial observability, while coordinating globally through value decomposition.
- **Scenario Engine with SUMO Adapter (Component 3):**
OpenCV and YOLOv8 process camera feeds to detect vehicles and emergency vehicles. Detected data is forwarded to SUMO in real time via TraCI, updating traffic flow and influencing agent behavior.
- **Real-time Evaluation Dashboard (Component 4):**
A driver-facing dashboard presents real-time road blockages or traffic redirections, generated from the MARL agents' output and historical blockage data.

The system simulates traffic control across multiple intersections while responding to real-world inputs. It combines simulation accuracy, learning adaptability, and human feedback output for real-world deployment potential.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

This project utilizes Multi-Agent Reinforcement Learning (MARL) under the Machine Learning specialization, aiming to improve urban traffic signal control, and will be demonstrated using the SUMO simulator. Each traffic intersection operates under an individual autonomous agent controlled through Deep Q-Networks (DQN) with decentralized execution. We hope to implement the Value Decomposition Network (VDN) architecture for agent coordination, allowing decomposition of global reward to individual agents' contributions without central control.

Required knowledge includes core reinforcement learning techniques like Q-Learning, DQN, and the multi-agent coordination approach of VDN. A MARL system must address non-stationarity arising from simultaneous multi-agent learning. This requires the design of reward structures and state spaces that facilitate decentralized coordination.

To enhance real-world relevance, the project incorporates real-time vehicle detection using camera feeds processed via OpenCV and YOLO. Detected vehicles will trigger live updates to the SUMO environment by generating vehicle instances and their attributes dynamically through the TraCI interface. This simulates smart intersection behavior by responding to real-world camera inputs.

Training data will include SUMO-generated vehicle states such as queue lengths, wait times, and signal statuses. Agent states will include local traffic metrics and neighboring signal information to support cooperative learning. The reward function will reflect both local performance (e.g., reduced waiting time) and global traffic efficiency.

Data requirements cover realistic traffic conditions including peak hours, random inflow, and emergency scenarios. Custom.rou.xml, .net.xml, and .sumocfg files will define network structure and dynamic vehicle routes.

The technology stack includes PyTorch for MARL algorithms, SUMO with TraCI for traffic simulation, and OpenCV with YOLOv5/YOLOv8 for vision-based vehicle detection. This demands expertise in RL algorithm development, vision-based object detection, real-time data integration, and multi-agent evaluation techniques.

8. Objectives and Novelty

Main Objective To design and implement a decentralized, intelligent traffic signal control system using Multi-Agent Reinforcement Learning (MARL) that optimizes urban traffic flow by dynamically adjusting signal timings based on real-time traffic conditions, while prioritizing emergency vehicle movement through integration with vision-based detection and the SUMO simulation environment.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Krushanth Amalanathan IT22920218	Formulate each traffic intersection's signal control as a Reinforcement Learning (RL) task by defining the agent's observation (state), available actions, and a reward function aligned with traffic efficiency goals.	1. Define agent state features including queue lengths, wait times, and neighbor signals. 2. Specify signal control actions (e.g., phase switching, timing adjustments). 3. Design a reward function balancing delay reduction and traffic throughput. 4. Formulate and validate the MDP model using simple intersection simulations.	Introduce an enhanced state and reward design that not only uses local traffic data (queues, wait times) but also incorporates neighboring intersection cues in the agent's observations. The reward function will be carefully shaped (e.g. heavy penalties for long vehicle waits) to encourage coordinated traffic flow, going beyond conventional single-intersection RL formulations which often ignore neighbor information.
Fernandopulle S N IT22000576	To design and implement a decentralized MARL system using VDN that	1.Design Inter-Agent Communication Protocol defining which state features agents will share.	Integrating inter-agent communication strategies within a VDN-based MARL system to improve decentralized traffic signal control

	allows inter-agent communication to improve coordination between neighboring intersections and reduce overall congestion through cooperative decision-making	2. Define Agent Architecture for VDN by Implement the value decomposition logic. 3. Implement VDN Training Loop with use of centralized critic with shared reward 4. Integrate Communication into Execution to ensure agents operate with local state + communicated neighbor data.	without relying on full global observability during execution.
Sivananthan Sangeerthan IT22346568	To design and develop a realistic, event-driven traffic simulation using SUMO that integrates both audio and visual emergency vehicle detection, trained specifically on Sri Lankan emergency vehicle patterns. The simulation enables dynamic and adaptive signal control to prioritize emergency vehicles, allowing MARL agents to be trained in	1. Audio-Based Emergency Detection A microphone-based detection system listens for emergency sirens in the vicinity. When detected, it flags a potential emergency event and activates the vision detection module. 2. Vision-Based Vehicle Confirmation (YOLO + OpenCV) Using real-time video streams, the system verifies the presence and type of vehicle (e.g., ambulance, fire truck). Once confirmed, it triggers the simulation response. 3. SUMO Network Configuration The simulation includes a detailed urban road	This component introduces a multi-modal sensing approach merging real-time audio cues (siren detection) with vision-based vehicle detection based on Sri Lanka's vehicle to simulate and prioritize emergency scenarios in SUMO. The audio module detects local siren sounds, while a YOLO-based vision model pre-trained on Sri Lankan ambulances, police vehicles, and fire trucks confirms the vehicle type via live camera feeds. When a siren is detected within a certain audio radius, the system activates cameras to verify the emergency vehicle type and location using YOLO. Once validated, a dynamic zone is

	environments that closely resemble local real-world conditions. Together, these modalities enable accurate, event-driven traffic management and allow MARL agents to be trained and evaluated in realistic, unpredictable urban scenarios.	network with multiple intersections, lane logic, and adaptive traffic light controllers, built using SUMO. 4. Dynamic Zone and Triggering System Upon verified detection, a virtual “priority zone” is created around the emergency vehicle. When the vehicle enters this zone, the traffic signal at the upcoming intersection is dynamically triggered to green, clearing the path. 5. Vehicle Spawn and Event Handling via TraCI Both normal and emergency vehicles are spawned in the simulation using the TraCI API, based on either patterned scenarios (rush hour, random) or detection events. 6. Emergency Vehicle Modeling Emergency vehicles are given high-priority routing, dynamic signal preemption, and are observable to agents through extended environmental states, allowing agents to learn to respond appropriately.	established, and upon vehicle entry, traffic signals ahead are pre-emptively triggered to turn green, allowing the vehicle uninterrupted passage. This simulation of real-world urgency enhances the realism of agent training.
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		7. Simulation Scenarios and Agent Training The system simulates diverse traffic conditions, such as normal flow, peak congestion, and emergency interventions. MARL agents are integrated with the simulation for live training and performance testing under stress and unpredictability.	
Luvinson I IT22323316	The primary objective of this component is to simulate realistic pedestrian behavior within a traffic environment by considering varying age groups such as children, adults, and elderly individuals, each with distinct mobility characteristics. It aims to develop an adaptive pedestrian signal control system that adjusts crossing durations in real-time	• Model pedestrian behavior in SUMO, including different walking speeds and crossing behaviors. • Simulate pedestrian detection through virtual buttons or presence triggers. • Build a dynamic pedestrian signal controller that adjusts green time based on crowd size and pedestrian type. • Integrate emergency vehicle detection to delay walk signals when necessary. • Simulate assistive technologies such as: ○ Countdown indicators	The novelty of this component lies in its human-centric, adaptive approach to pedestrian traffic management — a dimension often overlooked in MARL-based or vehicle-focused traffic systems. Unlike conventional pedestrian signal systems, this module introduces age-sensitive pedestrian modeling, where individuals are simulated with varying mobility characteristics (e.g., children, adults, and elderly pedestrians), each with different crossing speeds and needs. The system intelligently adjusts signal timings based on the pedestrian's age group, ensuring safer and more accessible crossings,

	based on pedestrian presence and walking speed. Additionally, the component seeks to integrate emergency vehicle awareness by temporarily pausing pedestrian signals during high-priority vehicle movements to ensure safety. To enhance accessibility, it includes assistive crossing features like visual countdowns and audio signals tailored for differently-abled or elderly pedestrians. Finally, it evaluates the overall effectiveness of the system by analyzing its impact on pedestrian safety, waiting time, and traffic flow, and compares it against	• Visual or audio signals for crossing safety • Evaluate key metrics: • Pedestrian wait time and safety rate • Vehicle delay due to pedestrian priority • Comparison against fixed pedestrian signal control	especially for slower-moving individuals like the elderly. Furthermore, it incorporates emergency-awareness, assistive technologies (like audio cues and visual countdowns), and operates independently of the learning agent. By doing so, it not only enhances the inclusivity and realism of the simulation but also supports smart city goals that prioritize safety and accessibility for all road users — not just vehicles.
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	traditional static signal control mechanisms		
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9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Krushanth Amalanathan IT22920218	This component lays the foundation by modeling each traffic light controller as an RL agent. We will precisely define what each agent “sees” as its state – for example, the number of vehicles or queue length on each incoming lane, the cumulative waiting time of those vehicles, the current signal phase, and even limited information about adjacent intersections’ signals. The agent’s action space will be specified, such as choosing which signal phase to switch to or when to switch (for instance, toggling green light between North-South and East-West directions after a minimum green interval). A core part of this formulation is the reward function design: the agent will receive feedback that captures traffic performance, e.g. a negative reward proportional to total waiting time or queue length at the intersection each timestep (penalizing long waits and encouraging flow). By defining the traffic signal control problem as a Markov Decision Process (MDP) with these states, actions, and rewards, we enable each agent to learn policies that minimize congestion and vehicle delay at its intersection. This formalization will be uniformly applied across all intersections, ensuring the multi-agent system has a consistent framework for learning and collaboration.
Fernandopulle S N IT22000576	This component focuses on implementing a Multi-Agent Reinforcement Learning (MARL) strategy where each agent trained using centralized training with decentralized execution (CTDE), using the Value Decomposition Network (VDN) architecture. This enable agents to collaborate through lightweight inter-agent communication to optimize citywide traffic flow. During training, agents receive a shared global reward, which is decomposed into individual agent value functions by VDN. Once trained, each agent makes decisions independently using local observations and limited exchanged signals from neighboring intersections. It will interface with the SUMO simulation engine to test the trained agents in realistic traffic patterns.
Sivananthan Sangeerthan IT22346568	This component introduces a multi-modal sensing approach merging real-time audio cues (siren detection) with vision-based vehicle detection based on Sri Lanka’s vehicle to simulate and prioritize emergency scenarios in SUMO. This module builds a lifelike urban traffic simulation environment for developing and testing multi-agent reinforcement learning (MARL) agents. The environment replicates real-world complexity using SUMO for traffic flow and integrates real-time emergency detection via both sound and video feeds. This simulation of real-world urgency enhances the realism of agent training.

Luvinson I IT22323316	This component focuses on improving pedestrian safety and accessibility in smart traffic systems. It simulates realistic pedestrian movement using SUMO and introduces a dynamic signal control system that adjusts crossing time based on real-time pedestrian presence and behavior. The system accounts for diverse pedestrian types, including elderly and differently abled individuals, and handles emergency scenarios by pausing pedestrian signals when emergency vehicles approach. Assistive features such as countdowns and audio cues are incorporated to enhance inclusivity. The component works independently of the MARL agent, contributing a human-centered layer to the overall traffic control system and increasing its deployability in real-world smart city contexts.
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10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Prof	Nuwan	Kodagoda	 2025/07/04
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

- a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	
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- b) Does the proposed topic exhibit novelty?

Yes		No	
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- c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	
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- d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	
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- e) Supervisor's Evaluation and Recommendation for the Research topic:

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Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.